```
import pandas as pd

file_path = "AutoEIT Sample Transcriptions for Scoring.xlsx"
# Load the first part
df = pd.read_excel(file_path, sheet_name="38001-1A")

df.head()
```

	Sentence	Stimulus	Transcription Rater 1	Score
0	1	Quiero cortarme el pelo (7)	Quiero cortarme el pelo	NaN
1	2	El libro está en la mesa (7)	El libro está en la mesa	NaN
2	3	El carro lo tiene Pedro (8)	El carro lo tiene Pedro	NaN
3	4	El se ducha cada mañana (9)	El se ducha cada mañana	NaN
4	5	¿Qué dice usted que va a hacer hoy? (9)	Que dices ustedes se que van a hacer hoy?	NaN

Next steps: New interactive sheet

```
import re
import unicodedata

# ==========================================
# TEXT NORMALIZATION FUNCTIONS
# ==========================================

def remove_accents(text):
    """Removes accent marks so 'películas' becomes 'peliculas' for fair comparison."""
    if pd.isna(text): return ""
    return ''.join(c for c in unicodedata.normalize('NFD', str(text)) if unicodedata.category(c) != 'Mn')

def clean_stimulus(text):
    """Removes syllable counts and normalizes target sentences."""
    if pd.isna(text): return ""
    text = str(text)
    text = re.sub(r'\(\d+\)', '', text)  # Remove syllable counts like (12)
    text = remove_accents(text)          # ACCENT FIX APPLIED
    text = re.sub(r'[^\w\s]', '', text)  # Remove punctuation
    return text.strip().lower()

def clean_transcription(text):
    """Removes human rater noise, hesitations, and false starts."""
    if pd.isna(text): return ""
    text = str(text)
    # Ignores hesitations and false starts per Exception rules
    text = re.sub(r'\[pause\]|\[gibberish\]|\[cough\]|xxx|um|uh', '', text, flags=re.IGNORECASE)
    text = remove_accents(text)          # ACCENT FIX APPLIED
    text = re.sub(r'[^\w\s]', '', text)  # Remove punctuation
    text = re.sub(r'\s+', ' ', text)     # Remove extra spaces
    return text.strip().lower()

# Apply the cleaning functions
df['Clean_Stimulus'] = df['Stimulus'].apply(clean_stimulus)
df['Clean_Transcription'] = df['Transcription Rater 1'].apply(clean_transcription)

# Verify the cleaning
df[['Clean_Stimulus', 'Clean_Transcription']].head()
```

	Clean_Stimulus	Clean_Transcription
0	quiero cortarme el pelo	quiero cortarme el pelo
1	el libro esta en la mesa	el libro esta en la mesa
2	el carro lo tiene pedro	el carro lo tiene pedro
3	el se ducha cada manana	el se ducha cada manana
4	que dice usted que va a hacer hoy	que dices ustedes se que van a hacer hoy

```
from sentence_transformers import SentenceTransformer, util

# ==========================================
# MEANING-BASED SCORING ENGINE
# ==========================================

print("Loading ML Semantic Model...")
ml_model = SentenceTransformer('paraphrase-multilingual-MiniLM-L12-v2')
```

```
print("Model Loaded!")

# Define Spanish function words to isolate "Content Words" (Idea Units)
FUNCTION_WORDS = {"el", "la", "los", "las", "un", "una", "unos", "unas", "de", "en", "a", "que", "y", "o", "pero", "se", "r

def calculate_score_strict(row):
    stimulus_raw = str(row['Clean_Stimulus']).strip()
    transcription_raw = str(row['Clean_Transcription']).strip()

    stimulus = stimulus_raw.split()
    transcription = transcription_raw.split()

    # TABLE 1: Score 0 Logic
    if not transcription or transcription_raw == 'xxx':
        return 0

    content_words_trans = [w for w in transcription if w not in FUNCTION_WORDS]
    if len(transcription) <= 2 or len(content_words_trans) <= 1:
        return 0

    # TABLE 5: Score 4 Logic
    if stimulus_raw == transcription_raw:
        return 4

    # Calculate Idea Units (Overlap) and Meaning (Semantic Score)
    stim_set = set(stimulus)
    trans_set = set(transcription)

    overlap_ratio = len(stim_set & trans_set) / len(stim_set) if stim_set else 0

    emb_stim = ml_model.encode(stimulus_raw, convert_to_tensor=True)
    emb_trans = ml_model.encode(transcription_raw, convert_to_tensor=True)
    semantic_score = util.cos_sim(emb_stim, emb_trans).item()

    # TABLE 4: Score 3 Logic
    if semantic_score >= 0.88 and overlap_ratio >= 0.70:
        return 3

    # TABLE 3: Score 2 Logic
    if overlap_ratio > 0.50 and semantic_score >= 0.60:
        return 2

    # TABLE 2: Score 1 Logic
    if overlap_ratio <= 0.50 or semantic_score < 0.60:
        return 1

    return 0

# Apply the  scoring to your test dataframe to see if it works
df['Automated_Score'] = df.apply(calculate_score_strict, axis=1)
df[['Clean_Stimulus', 'Clean_Transcription', 'Automated_Score']].head(15)
```

```
Loading ML Semantic Model...
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), s
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
modules.json: 100%  229/229 [00:00<00:00, 10.7kB/s]
config_sentence_transformers.json: 100%  122/122 [00:00<00:00, 5.56kB/s]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fa
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN
README.md: 3.89k/? [00:00<00:00, 169kB/s]
sentence_bert_config.json: 100%  53.0/53.0 [00:00<00:00, 3.39kB/s]
config.json: 100%  645/645 [00:00<00:00, 44.3kB/s]
model.safetensors: 100%  471M/471M [00:04<00:00, 160MB/s]
Loading weights: 100%  199/199 [00:00<00:00, 471.49it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100%  526/526 [00:00<00:00, 22.2kB/s]
tokenizer.json: 100%  9.08M/9.08M [00:00<00:00, 38.0MB/s]
special_tokens_map.json: 100%  239/239 [00:00<00:00, 10.8kB/s]
config.json: 100%  190/190 [00:00<00:00, 17.4kB/s]
Model Loaded!
```

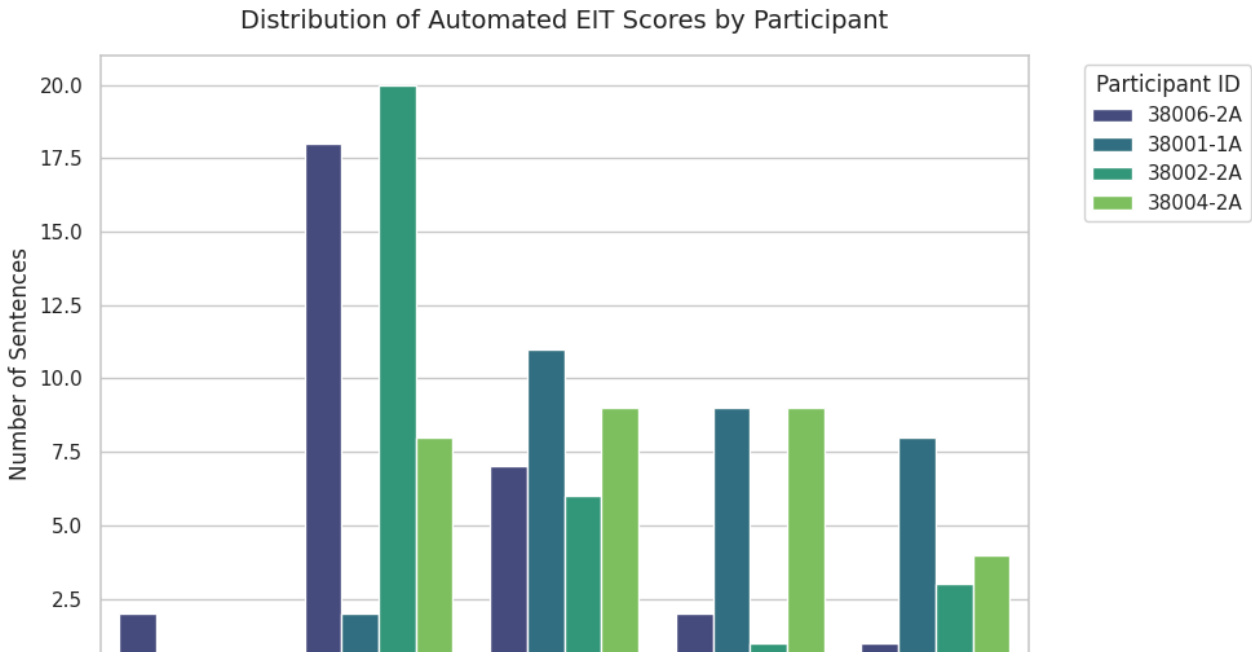
	Clean_Stimulus	Clean_Transcription	Automated_Score
0	quiero cortarme el pelo	quiero cortarme el pelo	4
1	el libro esta en la mesa	el libro esta en la mesa	4
2	el carro lo tiene pedro	el carro lo tiene pedro	4
3	el se ducha cada manana	el se ducha cada manana	4
4	que dice usted que va a hacer hoy	que dices ustedes se que van a hacer hoy	2
5	dudo que sepa manejar muy bien	dudo que sepa manajar bien	2
6	las calles de esta ciudad son muy anchas	las calles de esta cuidad son muy anchas	3
7	puede que llueva manana todo el dia	puede que lleva manana todo el dia	2
8	las casas son muy bonitas pero caras	las casas son muy bonitas pero muy cadas	2
9	me gustan las peliculas que acaban bien	me gustan las peliculas que acaban bien	4
10	el chico con el que yo salgo es espanol	el chico con yo salgo esta bien	2
11	despues de cenar me fui a dormir tranquilo	despues de cenar me fui a dormir tranquilo	4
12	quiero una casa en la que vivan mis animales	quiero una casa que vivan los animales	2
13	a nosotros nos fascinan las fiestas grandiosas	a nosotros nos fascina los fiestas grandiosos	2
14	ella solo bebe cerveza y no come nada	ella solo bebe cerveza y no come nada	4

```
# ==========================================
# EXECUTE, PRINT & EXPORT ALL SHEETS
# ==========================================

def process_all_participants(input_file, output_file):
    print(f"Loading workbook: {input_file}\n")

    # Read all sheets into a dictionary of DataFrames
    all_sheets = pd.read_excel(input_file, sheet_name=None)

    # Create an Excel writer to save multiple tabs into one output file
    with pd.ExcelWriter(output_file, engine='openpyxl') as writer:

        for sheet_name, df_sheet in all_sheets.items():
            # Skip the 'Info' tab or any tab without the correct columns
            if 'Stimulus' not in df_sheet.columns or 'Transcription Rater 1' not in df_sheet.columns:
                print(f"Skipping sheet: '{sheet_name}' (No transcript data found)\n")
                continue
```

```
            print(f"Scoring participant: '{sheet_name}'...")

            # Apply cleaning
            df_sheet['Clean_Stimulus'] = df_sheet['Stimulus'].apply(clean_stimulus)
            df_sheet['Clean_Transcription'] = df_sheet['Transcription Rater 1'].apply(clean_transcription)

            # Apply scoring
            df_sheet['Automated_Score'] = df_sheet.apply(calculate_score_strict, axis=1)

            # Format output
            final_output = df_sheet[['Sentence', 'Stimulus', 'Transcription Rater 1', 'Automated_Score']]

            # Save this participant's data to a new tab in the final Excel file
            final_output.to_excel(writer, sheet_name=sheet_name, index=False)

            # --- NEW: PRINT THE RESULTS FOR THIS TAB ---
            print(f"\n{'='*70}")
            print(f" PREVIEW FOR SHEET: {sheet_name}")
            print(f"{'='*70}")
            # Printing the first 15 rows of each sheet for a clean visual check
            print(final_output.head(15).to_string(index=False))
            print("\n")

    print(f" Success! All sheets scored and exported to: {output_file}")

# --- RUN THE CODE ---

INPUT_FILENAME = "AutoEIT Sample Transcriptions for Scoring.xlsx"
OUTPUT_FILENAME = "Final_Master_Scores_AutoEIT.xlsx"

process_all_participants(INPUT_FILENAME, OUTPUT_FILENAME)
```

```
 8           Puede que llueva mañana todo el día (12)           Puede que mañana todo la día                2
 9          Las casas son muy bonitas pero caras (12)    Las casas muy bonitos pero ... casas                2
10        Me gustan las películas que acaban bien (12)     Me gustan las películas campan bien                3
11        El chico con el que yo salgo es español (13)                el chico con ... español                1
12     Después de cenar me fui a dormir tranquilo (13)      después de finar ... él dormir xxx                1
13   Quiero una casa en la que vivan mis animales (14)           Quiere una casa con animales                1
14 A nosotros nos fascinan las fiestas grandiosas (14) a nosotros fana-xxx fiestas grandinosas                1
15          Ella sólo bebe cerveza y no come nada (15)                Ella xxx no beber querer                1


Scoring participant: '38004-2A'...

======================================================================
 PREVIEW FOR SHEET: 38004-2A
======================================================================
```

```
Success! All sheets scored and exported to: Final_Master_Scores_AutoEIT.xlsx
```

```
import matplotlib.pyplot as plt
import seaborn as sns

# ==========================================
# BLOCK 4: DATA VISUALIZATION
# ==========================================

def visualize_results(master_excel_file):
    print(f"Generating performance visualization from: {master_excel_file}...")

    # Read all sheets from the master file we just created
    all_data = pd.read_excel(master_excel_file, sheet_name=None)

    # Combine them into one dataframe for plotting
    combined_df = pd.DataFrame()
    for sheet_name, df_sheet in all_data.items():
        df_sheet['Participant'] = sheet_name
        combined_df = pd.concat([combined_df, df_sheet])

    # Set the visual style
    sns.set_theme(style="whitegrid")
    plt.figure(figsize=(10, 6))

    # Create a grouped bar chart
    ax = sns.countplot(
        data=combined_df,
        x='Automated_Score',
        hue='Participant',
        palette='viridis'
    )

    # Add labels and titles
    plt.title('Distribution of Automated EIT Scores by Participant', fontsize=14, pad=15)
    plt.xlabel('Rubric Score (0 - 4)', fontsize=12)
    plt.ylabel('Number of Sentences', fontsize=12)
    plt.legend(title='Participant ID', bbox_to_anchor=(1.05, 1), loc='upper left')

    # Save the chart as a image
    plt.tight_layout()
    image_filename = 'EIT_Score_Distribution.png'
    plt.savefig(image_filename, dpi=300)


    plt.show()
    print(f" Visualization saved as: {image_filename}")

# --- RUN THE VISUALIZER ---
MASTER_FILE = "Final_Master_Scores_AutoEIT.xlsx"
visualize_results(MASTER_FILE)
```

```
Generating performance visualization from: Final_Master_Scores_AutoEIT.xlsx...
```